

B.Tech. Degree V Semester Examination November 2017

ME 15-1506 POWER PLANT ENGINEERING (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define load factor and demand factor.
 (b) Which types of power plants are used as base load power plants? Why are such power plants used as base load power plants?
 (c) What is a Surge tank? What is its function?
 (d) Explain the thermodynamic cycle of a simple gas turbine using T-s diagram.
 (e) Why diesel engines are heavier than petrol engines for the same power output? What is the range of compression ratios of diesel engines?
 (f) What are the advantages of gas turbines over diesel engines when used in power plants?
 (g) List any four types of Nuclear reactors.
 (h) What are the factors that determine the selection of methods for coal burning?
 (i) What is solar constant? What is its average value? What is solar insolation?
 (j) Explain the principle of ocean thermal energy conversion system.

PART B

(4 × 10 = 40)

- II. (a) A power station has to meet the following load demand: (5)
 Load A-50 kW between 10 am and 6 pm. Load B-30 kW between 6 pm and 10 pm. Load C-20 kW between 4 pm and 10 am.
 Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.
 (b) At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected loads are as under: (5)

Transformer	Load	Demand factor	Diversity of groups
Transformer No.1	12 kW	0.65	1.5
Transformer No.2	14 kW	0.6	3.5
Transformer No.3	16 kW	0.7	1.5

If the diversity factor among the transformers is 1.35, find the maximum load on the feeder.

OR

(P.T.O.)

- III. (a) The annual peak load on a 30 MW power station is 25 MW. The power station supplies load having maximum demands of 10 MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 0.45. Find: (5)
- (i) Average load.
- (ii) Diversity factor.
- (iii) Demand factor.
- (b) In a steam power plant the capital cost of power generation equipment is ₹2500000. The useful life of the plant is 25 years and salvage value of the plant is ₹100000. Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 5%. (5)
- IV. (a) What are the starting methods used in large diesel engine power plants? Explain any one method. (5)
- (b) Draw the layout of a gas turbine power plant and explain its working. (5)
- OR**
- V. (a) Derive the equation for the thermal efficiency of a simple diesel engine cycle. (5)
- (b) What are the methods used for improving the thermal efficiency of a gas turbine? Explain any one method. (5)
- VI. (a) Draw the block diagram of a steam power plant and explain its working. (5)
- (b) What are the various components of a nuclear reactor? Explain the function of each. (5)
- OR**
- VII. (a) Flue gases passing through a chimney of a steam power plant is 300°C. Calculate the mass of flue gas when the draught produced is equal to 2 m of water, when the flue gas formed is 23 kg per kg of fuel burnt. The diameter of the chimney is 2 meters. (5)
- (b) Explain the different cooling methods used in nuclear reactors. (5)
- VIII. (a) Derive Betz equation to calculate the theoretical efficiency of wind turbines. What are the various types of losses that decrease the efficiency of wind turbine? What is the approximate net efficiency of a wind turbine? (5)
- (b) Explain the working of two types of solar collectors with neat sketches. (5)
- OR**
- IX. (a) Describe the working of an open cycle MHD system with a sketch. (5)
- (b) Explain the working of a geothermal power plant. (5)
